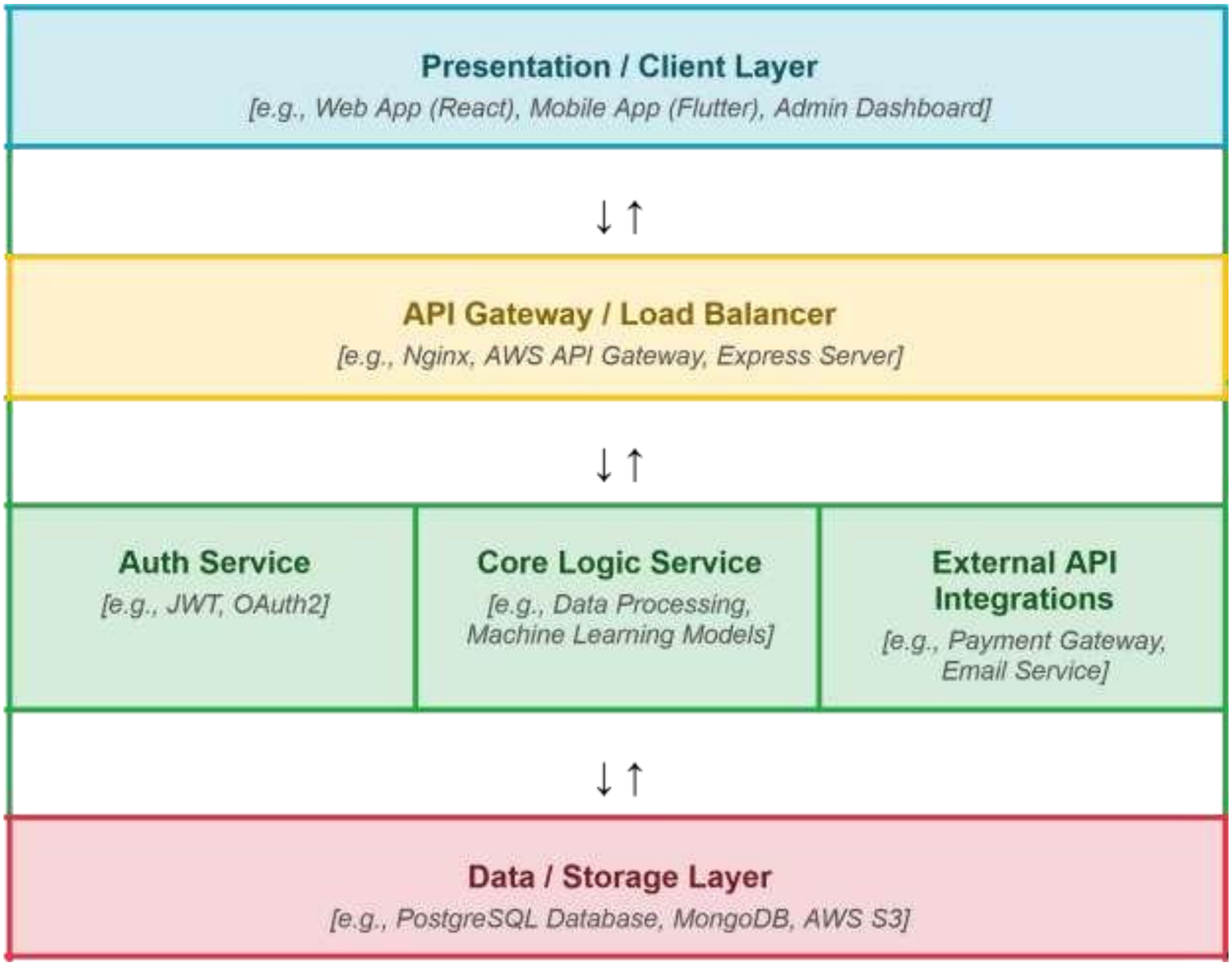
Solution Architecture

Date	23 JUNE 2026
Team ID	SWTID-2026-7169
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:
Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Renders the Home, About, and Find Your Crop pages; collects user input (N, P, K, temperature, humidity, pH, rainfall) via a form and displays the predicted crop.	HTML, CSS, Bootstrap, Jinja2 (Flask templating)
API Gateway	Not used in current architecture — Flask's built-in routing handles all requests directly (Home, About, FindYourCrop, Predict routes)	Flask routing (@app.route)

Microservices	Single Flask service that receives form input, passes it to the trained ML model via model.predict(), and returns the crop recommendation to the UI	Python, Flask, Scikit-learn, Pickle
Database	Stores the training dataset used to build the model and the serialized model file used for inference; no live database/transactional storage is used	CSV file (Crop_recommendation.csv), Pickle (model.pkl)